Weifan Lu

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Research Interests

I am an observational seismologist whose research aims to improve our understanding of seismic source processes, including fast and slow earthquakes, volcanic processes, and their interactions and triggering mechanisms. My research combines the development of novel methods, the analysis of large datasets, and onshore geophysical data collection.

Keywords: Earthquake Physics, Earthquake Detection, Slow Earthquakes, Earthquake Interaction and Triggering, Volcanic Seismology, Waveform Modeling, Subduction Zones, Tidal Modulation, Machine Learning

Academic Positions

Université Grenoble Alpes, Grenoble, France 2025.12 – Present
CNRS Postdoctoral Fellow
Advisor: Nikolai Shapiro

The University of Tokyo, Tokyo, Japan 2025.07 – 2025.12
Postdoctoral Fellow
Advisor: Satoshi Ide

Education

The University of Tokyo, Visiting Ph.D. Student in Geophysics 2023 – 2025
Advisor: Satoshi Ide

Peking University, Ph.D. in Geophysics 2019 – 2025
Advisors: Han Yue & Lian Xue

Awards and Honors

European Commission's Seal of Excellence, European Commission 2026
JSPS Postdoctoral Fellowship, Japan 2025
Merit Student Award, Peking University 2022, 2023
Tung OOCL Scholarship, Peking University 2023
President Scholarship, Peking University 2022
Outstanding Scientific Research Award, Peking University 2021
Outstanding Presentation, China Seismic Experimental Site Conference 2020
National Scholarship, Ministry of Education of the People's Republic of China 2018
Mathematical Contest in Modeling Certificate of Achievement, Meritorious Winner (top 10%) 2018
Liu Guang-Ding Geophysics Scholarship, Liu Guang-Ding Science Foundation 2017
National Scholarship, Ministry of Education of the People's Republic of China 2016

Publications

Under Review or In Revision

[S1] **Lu, W.**[*], Shapiro, N., Melnik, O., Lyakhovsky, V., Münchmeyer, J., Journeau, C., Soubestre, J., & Seydoux, L. Deep seismo-volcanic tremors constrain the magma supply from the mantle. *Under Review at Science*. Preprint at EarthArXiv. doi: 10.31223/X5HJ77.

[S2] **Lu, W.*** TiMEpy: A Python Package for Analyzing Tidal Modulation of Fast and Slow Earthquakes. *Under Review*. Preprint at EarthArXiv. doi: 10.31223/X5P77P.

Peer Reviewed

[1] **Lu, W.***, Ide, S., & Ren, C. (2026). Abrupt Suppression of Tectonic Tremors Outside the Stress Shadow of Mendocino Earthquakes. *Geophysical Research Letters*. doi:10.1029/2026GL123699.

[2] **Lu, W.***, Shapiro, N., & Münchmeyer, J. (2026). Magma Pathways Beneath the Klyuchevskoy Volcanic Group, Kamchatka, Revealed by a Machine-Learning-Based Earthquake Catalog. *Geophysical Research Letters*. doi:10.1029/2026GL123629.

[3] **Lu, W.***, & Ide, S. (2026). Spatiotemporal Characteristics of Tectonic Tremor in California. *Geophysical Research Letters*. doi:10.1029/2025GL120742.

[4] **Lu, W.*** (2026). Migration of Multimodal Deep Earthquake Swarm beneath the Abu Volcano Group, Japan. *Geophysical Research Letters*. doi:10.1029/2025GL119316.

[5] Yue, H.*, **Lu, W.**, & Li, G. (2026). Post-foreshock transient slip facilitates the mainshock occurring during the 2016 Kumamoto earthquake sequence: the fourth foreshock evolution mode. *Earth and Planetary Science Letters*. doi:10.1016/j.epsl.2026.119905.

[6] Chen, Q., Wang, L., Xue, L.*, **Lu, W.**, Li, Y.*, Zhao, L., & Wu, Q. (2026). Analysis of seismic cluster distribution and focal mechanism characteristics in Changning shale gas field, Sichuan. *Chinese Journal of Geophysics* (in Chinese). doi:10.6038/cjg2025T0392.

[7] **Lu, W.**, Xue, L.*, Yue, H., Zhuang, J., & Zhao, L. (2025). Exploring Tidal Modulation of Seismicity in Southern California. *Journal of Geophysical Research: Solid Earth*. doi:10.1029/2025JB032249.

[8] **Lu, W.***, Ide, S., & Yue, H. (2025). Feature of cascading rupture frequently observed in Northern California. *Communications Earth & Environment*. doi:10.1038/s43247-025-02138-y. (Behind the Paper)

[9] **Lu, W.***, Ide, S., & Yue, H. (2025). Small-scale stress heterogeneity inferred many Anti-repeating earthquakes in Sierra Valley, Nevada. *Earth, Planets and Space*. doi:10.1186/s40623-025-02273-y.

[10] Zhao, Z., Xue, L.*, Burgmann, R., Heimisson, E., **Lu, W.**, & Yue, H. (2025). Periodic seismicity modulations reveal important roles of pore fluid behaviors in controlling fast and slow earthquakes. *Science Advances*. doi:10.1126/sciadv.ady6350.

[11] **Lu, W.**, Zhao, Z., Yue, H.*, Zhou, S., Wu, J., & Song, X. (2024). The Accessible Seismological Dataset of the high-density 2D seismic Array along Anninghe Fault. *Earthquake Science*. doi:10.1016/j.eqs.2023.11.001. (Article Highlights)

[12] **Lu, W.**, Zhou, Y., Zhao, Z., Yue, H.*, & Zhou, S. (2022). Aftershock sequence of the 2017 Mw 6.5 Jiuzhaigou, China earthquake monitored by an AsA network and its implication to fault structures and strength. *Geophysical Journal International*. doi:10.1093/gji/ggab443. (PKU SESS News)

[13] Yue, H.*, Shen, Z.-K., Zhao, Z., Wang, T., Cao, B., Li, Z., Bao, X., Zhao, L., Song, X., Ge, Z., Ren, C., **Lu, W.**, Zhang, Y., Liu-Zeng, J., Wang, M., Huang, Q., Zhou, S., & Xue, L. (2022). Rupture process of the 2021 M7.4 Maduo earthquake and implication for deformation mode of the Songpan-Ganzi terrane in Tibetan Plateau. *Proceedings of the National Academy of Sciences*. doi:10.1073/pnas.2116445119.

[14] Chang, Y., Liang, C.*, Cao, F., Zhou, L., Liao, J., **Lu, W.**, & Wang, C. (2022). Seismic Velocity Structure for the Anninghe - Zemuhe Fault Zone by Wave Gradiometry Analysis. *Chinese Journal of Geophysics* (in Chinese). doi:10.6038/cjg2022P0731.

[15] Yue, H.*, Sun, J., Wang, M., Shen, Z., Li, M., Xue, L., **Lu, W.**, Zhou, Y., Ren, C., & Lay, T. (2021). The 2019 Ridgecrest, California earthquake sequence: Evolution of seismic and aseismic slip on an orthogonal fault system. *Earth and Planetary Science Letters*. doi:10.1016/j.epsl.2021.117066.

Selected Conference Presentations

[1] **Lu, W.**, Shapiro, N., & Münchmeyer, J. Dense Array and Machine Learning Reveal Detailed Relationship between Seismicity and Volcano Magmatic Activity beneath Klyuchevskoy Volcanic Group, Kamchatka (Oral), *EGU Meeting*, Vienna, Austria, 2026.

[2] Ide, S., & **Lu, W.** Spatiotemporal Characteristics of Tectonic Tremor in California (Poster), *EGU Meeting*, Vienna, Austria, 2026.

[3] **Lu, W.**, Ide, S., & Yue, H. Small-scale Stress Heterogeneity Inferred Many Anti-Repeating Earthquakes in the Sierra Valley, Nevada (Oral), *International Joint Workshop on Slow-to-Fast Earthquakes*, Florence, Italy, 2026.

[4] **Lu, W.**, Ide, S., & Yue, H. Small-scale Stress Heterogeneity Inferred Many Anti-Repeating Earthquakes in the Sierra Valley, Nevada (Poster), *JpGU Meeting*, Chiba, Japan, 2025.

[5] **Lu, W.**, Ide, S., & Yue, H. Feature of cascading rupture frequently observed in Northern California (Poster), *AGU Fall Meeting*, Washington, D.C., USA, 2024.

[6] **Lu, W.**, Ide, S., & Yue, H. Feature of cascading rupture frequently observed in Northern California (Oral), *SSJ Meeting*, Niigata, Japan, 2024.

[7] **Lu, W.**, Ide, S., & Yue, H. Large and small earthquakes can have similar seismic wave onsets of the San Andreas fault in Northern California (Oral), *JpGU Meeting*, Chiba, Japan, 2024.

[8] **Lu, W.**, Xue, L., & Yue, H. Tidal Modulation Constrained Fault Properties: A Case Study of Southern California, USA (Poster), *AGU Fall Meeting*, San Francisco, USA, 2023.

[9] **Lu, W.**, Xue, L., & Yue, H. Tidal Triggering of Earthquakes in Southern California (Oral, Online), *AGU Fall Meeting*, Chicago, USA, 2022.

[10] **Lu, W.**, Xue, L., & Yue, H. Tidal modulation of seismicity in Southern California (Poster, Online), *AGU Fall Meeting*, New Orleans, USA, 2021.

[11] **Lu, W.**, Yue, H., & Zhou, Y. Aftershock sequence and fault geometry of the 2017 Mw 6.5 Jiuzhaigou earthquake: an application of AsA network, denoising algorithms and fault geometry determination techniques (Oral), *AGU Fall Meeting*, San Francisco, USA, 2019.

Seminars / Talks

EOS (**Invited**), Nanyang Technological University, Singapore	2026.08
Géoazur (**Invited**), Nice, France	2026.05
ISTerre (**Invited**), Grenoble, France	2026.04
Chinese Academy of Sciences (**Invited**), Online, China	2026.01
GNS Science, Wellington, New Zealand	2025.10
Monash University, Melbourne, Australia	2025.08
1st Mini-Workshop of the ANR PREMs Project, ENS (**Invited**), Online, France	2025.07
Yunnan University (**Invited**), Kunming, China	2025.06
Tokyo University of Science, Tokyo, Japan	2025.04
MIT, Boston, USA	2024.12
Kyoto University, Kyoto, Japan	2024.11
Youth Forum on Earthquake Science and Technology (**Invited**), Beijing, China	2022.10

Services

Reviewer

Journal of Geophysical Research: Solid Earth (6)	2024 – Present
Geophysical Research Letters (2)	2026 – Present

Journal of Geophysical Research: Machine Learning and Computation (1) 2026 – Present
Seismological Research Letters (1) 2022 – Present
Earth, Planets and Space (4) 2026 – Present
Scientific Data (2) 2026 – Present
Earthquake Science (3) 2025 – Present

Volunteer

Science of Slow to Fast Earthquakes (JpGU Meeting) 2024 – Present
Book: Memories of Professor Don Helmberger Master of Seismology and Mentor Extraordinaire 2023

Affiliations / Memberships

American Geophysical Union (AGU) 2019 – Present
Japan Geosciences Union (JpGU) 2024 – Present
Seismological Society of America (SSA) 2025 – Present
Seismological Society of Japan (SSJ) 2024 – Present

Teaching Experience

Teaching Assistant

Earthquake Observation Internship, The University of Tokyo Spring 2025
Geophysics Essay Writing in English, Peking University Spring 2021
Earthquake Observation Internship, Peking University 2018 – 2022

Research Mentoring

Peking University undergraduate student Yan Pei 2022 – 2023
Peking University visiting undergraduate student Lemeng Wang (University of Toronto) 2024

Software

TiMEpy: A Python Package for Detecting Tidal Modulation in Fast and Slow Earthquakes
Auto-FtCluster: Determine fault geometry using earthquake locations and focal mechanisms

Field Experience

Anninghe-Zemuhe Array, 161 short-period seismometers, lead field student 2020 – 2021
Longmenshan Array of Small Arrays, 50 short-period seismometers, lead field student 2018 – 2020